Countersign: catching impersonation scam calls on the phone

How one call is judged. Everything on this page runs on the protected user's phone; nothing is sent.

Team losers · RAKSHAM 2026 Finale · Track 1
github.com/dope-XD/countersign

2.3%
of genuine bank calls warned
ICFD-31k held out, n = 1,019; keyword baseline 50.1%

87.8%
of scam calls warned
ICFD-31k test split, n = 4,499; 59.9% reach red

75.8%
of unseen scam types warned
5 fraud types absent from training, n = 760

28 / 30
spot-check kit correct
10 calls, 10 messages, 10 voices

10 / 10
fresh scams caught
written after the system was frozen

1 Hear the call

Capture
- Accessibility service keeps the mic during a call
- 16 kHz mono, memory only, speakerphone

Segment: Silero VAD
- Threshold 0.5; 0.4 s silence ends a segment
- Segments up to 12 s; audio dropped after use

Transcribe: Omnilingual ASR
- Meta, 300M, CTC, int8, sherpa-onnx
- Chosen on our 59 calls: WER 28% vs 65% / 44% for IndicConformer Hindi / English

2 Collect evidence

Judge 1: conversation model
what the caller is doing
- multilingual-e5-small (118M) fine-tuned by us; mean pooling + logistic head
- Trained on 267,601 windows: ICFD-31k (synthetic bank calls, EN + Hinglish) and our sets ×8
- Reads the last 320 tokens of the call; fp32 ONNX, 470 MB (int8 drifted by 0.20)
- s = mean of the last two scores; amber at $s \geq 0.142$, red at $s \geq 0.9855$ twice
- Thresholds set on 1,788 genuine dev calls at 3% and 0.5% false positives

Judge 2: cue rules + move matcher
what was said, and why
- 19 categories in English, Hinglish, Hindi; negation-aware (“we never ask for OTP”)
- Amber: 2 strong categories; red: 4 incl. an action and a scam type
- Talk about scams never counts towards red
- 17 written scam moves matched by e5 embeddings give the reason shown

Phone signals
- Unknown or foreign number; AnyDesk / TeamViewer or a payment app opened; call over 15 min
- Caller check failed; the user reading out 4–8 digits on an unknown call

Voice-clone check
- First 6 layers of AntiDeepfake wav2vec + our head, int8, 127 MB; first 20 s
- Equal error rate 23–35% on phone audio, so it can lift to amber, never to red

3 Decide

Combiner
- Amber = the higher judge
- Cue words alone need Judge 1 at watch or above
- Red needs two independent sources: both judges, a judge plus a strong phone signal, the user, or the server
- The level never goes down in a call

4 Act

Amber card
- Names the scam and the reason
- “Alert family”, “I know this caller”

Second opinion
- Optional server after amber: ASR 1B, same model, 24-layer voice model
- Audio in memory; can only raise

Red
- “This is likely a scam. Hang up.”
- Family alarm (page 2)

Kotlin on the phone and Python on the server give identical verdicts on shared test vectors, checked in CI. One 798 MB model download; detection then works offline.

After the verdict: the family, privacy, and how well it works

The response path, what data goes where, and results on data the model never trained on.

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5 Reach the family at red

Scam alert to the relay

- Once per call; call id, level, scam type, cue ids
- No words, no audio, no numbers

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Family member's app online?

- Yes: full-screen alarm over the lock screen, alarm tone at full volume
- “Mummy may be on a scam call”, Call / Message
- No: SMS from the user's own SIM

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Acknowledged within 60 s?

- Yes: escalation stops
- No: text every member and call the first one from the user's phone

6 Is it really Rahul?

Caller check (no audio)

- Each family phone reports its own calls (member ids, never numbers)
- The relay looks for Rahul's outgoing call from 45 s before to 5 s after the ring; waits 6 s

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Verdict on the card

- Rahul's phone called: ✓ verified
- No call, his app seen in 2 min: ✗ “Not Rahul's phone”
- App unseen: “Couldn't check”, never a false ✗
- A cloned voice or spoofed number fails: Rahul's phone never placed the call

A false alarm costs two taps

- “I know this caller” or “It's not a scam”, then “Yes, it's fine”
- Warnings stop; Rahul sees “False alarm”; one ids-only record for review

7 Privacy by construction

Call audio	phone memory, dropped per segment; server only after amber, in memory
Evidence	one 20 s AES-256 clip for a confirmed scam, deleted after 30 days
Transcript	phone memory for the call; never stored or sent
Numbers	on the phone only, matched to member ids by the last 10 digits
Relay	member ids, check verdicts, level, scam type, cue ids

8 Measured on data it never trained on (warned / red, %)

Set	n	Countersign	Keyword baseline
ICFD genuine, clear	1,019	2.3 / 0.4	50.1 / 8.9
ICFD genuine, ambiguous	511	4.5 / 0.8	70.8 / 15.5
ICFD scam, test split	4,499	87.8 / 59.9	94.3 / 51.5
ICFD unseen types, clear	760	75.8 / 42.9	90.9 / 34.7
ICFD unseen types, subtle	240	54.6 / 17.5	78.3 / 22.5
Our team's scam calls	16	93.8	–
Adversarial rewrites (scam / genuine)	16 / 14	75.0 / 21.4	–

Median time to amber 36–45 s. The keyword baseline flags more scams only because it flags most genuine calls too.

9 Known limits, and what's next

Weak today

- Polite, scripted English scams
- People talking about a scam
- Genuine calls in a computer voice reach amber
- Earpiece: the caller is barely heard; needs speaker
- ICFD-31k is synthetic: needs a pilot on real calls

Next

- Full-screen stop over payment apps (designed; the card warns today)
- On-phone language model as a third judge (prompt written and tested)
- Bank pilot: the call-centre version already runs on our server